MOHAMMED ZAKARIA MUTLAQ

Data Engineer — Pipelines, Warehouses, Data Quality & AI Data

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SUMMARY

Data Engineer with 2 years building production batch and streaming pipelines, dimensional warehouses, and automated data quality controls in Python and SQL. Ships Airflow, dbt, and Spark pipelines over MySQL, PostgreSQL, and ClickHouse on Azure, AWS, GCP, and Huawei Cloud, and builds the data layer behind production AI (retrieval, recommendations, evaluation). Five systems live in production for employers and clients.

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, PySpark), SQL (CTEs, window functions, tuning), Bash, Scala
Pipelines & Streaming: Airflow, dbt, Apache Spark, Kafka, Flink, NiFi, Debezium CDC, REST APIs, web scraping (Playwright)
Warehousing & Databases: PostgreSQL, MySQL, SQL Server, ClickHouse, BigQuery, Delta Lake, Hive, Redis, star schema, SCD2
Data Quality: Great Expectations, dbt tests and contracts, reconciliation, schema-drift gates, pytest, CI gates
Cloud: Azure (Data Factory, Event Hubs, Key Vault), AWS (EC2, RDS, S3), GCP (Cloud Storage), Huawei Cloud, Microsoft Fabric
DevOps & BI: Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Linux, Power BI (DAX), Grafana
AI Data: RAG, embeddings, vector search (FAISS, Qdrant), LLM evaluation, recommender systems, MLflow, FastAPI

PROFESSIONAL EXPERIENCE

Data Engineer — Waffarha, Cairo *Jul 2026 – Present*

- Run a live nightly recommendation pipeline (Jenkins, Docker): MoEngage event exports from Google Cloud Storage, deduped user histories, item-to-item similarity, an active-offer filter from ClickHouse, and 5,000-row batch upserts into MySQL on AWS RDS.
- Own the data platform behind a bilingual (Arabic/English) customer service assistant: Airflow DAGs from MySQL and ClickHouse with content-hash dedup, SCD2 history, and change-triggered re-indexing; 87.8% retrieval hit rate over a 9,106-doc BGE-M3 index.
- Built a daily 6-stage partner price-integrity control (ingest, scrape, match, compare, report) that flags prices outside a 2% tolerance, with per-partner checkpoints and Excel and HTML exception reports.
- Redacted 160,944 real support chats into a 117,915-conversation fine-tuning set with zero PII leaks across 8 patterns; built a 225-case bilingual eval suite with pass/fail exit codes.

Freelance Data Engineer — Self-employed, Cairo/Remote *Jan 2026 – Present*

- Delivered a client clinic warehouse: 6-stage Excel-to-PostgreSQL ETL (preflight, schema-drift check, quality scan, diff, backup, load) into a star schema with bridge tables, plus MySQL loaders and Power BI design.
- Built crawlers that harvest contractor and tender datasets for a construction client, with enrichment, dedup, and validation.

Big Data Engineering Intern — National Telecommunication Institute (NTI), Smart Village *Apr 2026 – Jul 2026*

- Built Spark and SQL ETL on Hadoop (HDFS, Hive, HBase) with Airflow, Kafka, Flume, and Sqoop ingestion, and AWS runs.

Data Engineer — Nexlify Startup, Cairo *Jan 2025 – Jan 2026*

- Built Airflow intake from 10+ clinics (Excel, APIs) into a from-scratch star-schema warehouse with fact, dimension, and bridge tables; reporting cycles improved 40%.
- Refactored SQL with CTEs and tuned joins across PostgreSQL, SQL Server, and MySQL: 30% lower latency, zero BI timeouts.
- Cleansed and validated millions of mixed Arabic/English records in Python before load; one Power BI model serves all clinics.

PROJECTS

Governed Lakehouse (Estates Data Platform) *dbt, Airflow, Great Expectations, DuckDB, GitHub Actions*

- 8/8 sources reconciled with zero unexplained loss; 160 dbt tests, 21 contracts, and 133 YAML quality checks on Great Expectations and DuckDB (196 PASS, 0 ERROR); 7 Airflow DAGs with watermarks and 20 live demos.

Telecom Data Platform *PySpark, Delta Lake, dbt, scikit-learn, Terraform, pytest*

- Designed a medallion lakehouse for 8 telecom domains on synthetic data: SHA-256 ID hashing, Great Expectations rules for CDR, KPI, and billing, a schema-drift gate, and Azure Terraform; 80 pytest tests.

Real-Time Flight Tracking *Kafka, Flume, Spark Structured Streaming, InfluxDB, Grafana*

- Live ADS-B feed every 10 s via Flume and Kafka into Spark Structured Streaming, InfluxDB, HDFS, and Grafana.

EDUCATION, CERTIFICATIONS & LANGUAGES

B.Sc. in Computers and Artificial Intelligence — Helwan University, Cairo *Sep 2020 – May 2024*
GPA 3.3 | Graduation Project: A+ (Arabic hadith Q&A)

Huawei HCIP Big Data (H13-723), 2026 | AWS Certified Cloud Practitioner, 2026 | UZH Blockchain Deep Dive, 2023
Languages: Arabic (native), English (C1, professional)